6-10 Information Design

Point!

1 Information Design

(1) ([1]**Information design**) : The process of organizing and creatively expressing information according to its purpose, thereby ensuring that the intended message is correctly conveyed to the target audience.

(2) Information Design Methods

[1] ([2]**Abstraction**) : Conveying the intended information simply from within a large amount of data.

- ([3]**Pictogram**) : A pictorial symbol designed for the purpose of conveying information without using words. It is used for information signs at stations and airports.
- ([4]**Icon**) : A representation of a program's content on a computer or smartphone depicted through images or illustrations.



Pictogram



Icon

[2] ([5]**Visualization**) : Visually representing information to make it more understandable.

<Examples> Tables, graphs, etc.

[3] ([6]**Structuring**) : Organizing and expressing information clearly by arranging it according to relationships, connections, levels, stages, order, etc.

<Examples> Hierarchical menus on web pages, department store floor maps, etc.

2 Design Techniques for Ease of Understanding

(1) ([7]**User interface (UI)**) : A system for transferring information between humans and devices

<Examples> Voice input, touch panel, keyboard, etc.

[1] ([8]**CUI**) : A user interface where a device is operated by entering commands via a keyboard.

[2] ([9]**GUI**) : A user interface that allows users to issue commands in an intuitive and easy-to-understand way using icons and buttons.

(2) ([10]**User experience (UX)**) : The experience or emotional impact that users gain from interacting with a product or service.

(3) ([11]**Affordance**) : Actions or operations that can be performed on an object.

(4) ([12]**Signifier**) : Cues that prompt users to take action.

(5) ([13]**Usability**) : A measure used to indicate how easy and understandable it is for users.

(6) ([14]**Accessibility**) : The ease of access to information and services for a wide range of people.

(7) ([15]**Universal design**) : Design that is thoughtfully created with the intention that all people, regardless of age, language, nationality, or physical ability, can use it without difficulty.

3 Organizing and Classifying Information

([16]**LATCH method**) : A principle for organizing information and presenting it in a way that is easy for users to understand. Information is organized and classified according to the following five criteria.

- Location : Classification based on physical location.
- Alphabet : Classification based on alphabetical or syllabary order.
- Time : A classification based on the sequence of events in terms of time.
- Category : A classification based on the differences between things, used as a criterion to distinguish realms.
- Hierarchy : Classification based on quantitative changes such as size, level, etc.

Warm Up

Answer the following questions.

(1) Choose the most appropriate example of visualization as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Pictograms for each classroom were created to guide people around the school building.

B A web page to introduce the school was created. The page was organized with menus for each hierarchy.

C The survey results were compiled into tables and graphs.

D Text summarizing multiple matters is presented in bullet points.

(2) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

> Current computers widely use ([1]), which allows users to manipulate graphically displayed information, such as icons, with a mouse or their finger. On the other hand, early computers used ([2]), which was a method for conveying commands by inputting characters from a keyboard.

A CUI **B** GUI **C** VUI **D** NUI

(3) What is the term for a design that is thoughtfully created in advance to allow all individuals, regardless of nationality, gender, age, or physical ability, to use it without difficulty? Choose one from the options **A** to **D** below, and answer using the letter.

A Usability **B** Universal design

C Signifier **D** User interface

Explanation

(1) **A** is an example of abstraction, **B** of structuring, **C** of visualization, and **D** of structuring.
Therefore, **C**

(2) [1]: **B** [2]: **A**

(3) **B**

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Methods of information design include the following.
> ([1]): A method for discerning the overall picture and clearly organizing the relationships between elements.
> ([2]): A method for visually representing information.
> ([3]): A method for extracting only the necessary elements from a large amount of information.

A Specification **B** Abstraction **C** Visualization **D** Materialization **E** Structuring

(2) Choose the most appropriate example of abstraction as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Pictograms for each classroom were created to guide people around the school building.
B A web page to introduce the school was created. The page was organized with menus for each hierarchy.
C The survey results were compiled into tables and graphs.
D Text summarizing multiple matters is presented in bullet points.

(3) Choose one term that represents a pictorial symbol that conveys information by abstracting the intended message without using text from the options **A** to **D** below, and answer using the letter.

A Usability **B** Affordance
C Signifier **D** Pictogram

2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

> The experience that users gain through products or services is referred to as ([1]). The aim is not just to achieve ease of use, but also to achieve a sense of comfort. For that purpose, it is necessary to simplify the ([2]) that people use to operate devices and screens, choose colors with consideration for the fact that different people perceive colors differently, and add textual information.

A GUI **B** CUI **C** UX **D** UI

(2) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **F** below, and answer using the letters.

> Regardless of the level of advancement for functions offered by a computer, those functions cannot be effectively applied in real life if they are difficult to use. The ease of use of such devices is referred to as ([1]), and ([1]) varies depending on the user. On the other hand, ease of use for a wide range of people, regardless of their abilities or conditions, is referred to as ([2]). The term ([3]) refers to design that enables ease of use for everyone, regardless of age, language, nationality, or physical ability.

A Accessibility **B** Usability **C** Affordance
D Universal design **E** User interface **F** Barrier-free

Exercise

1 Answer the following questions.

(1) Answer the following questions.

[1] Choose the most appropriate explanation of visualization as a method used in information design from the options **A** to **D** below, and answer using the letter.

A Extracting only the necessary elements from a large amount of information.

B Discerning the overall picture and clearly organizing the relationships between elements.

C Expressing the information you want to convey in a simple manner using diagrams or illustrations.

D Representing information visually.

[2] Choose all the appropriate examples of structure as a method used in information design from the options **A** to **D** below, and answer using the letters.

A A web page to introduce the school was created. The page was organized with menus for each hierarchy.

B The survey results were summarized into a graph to convey information in an easy-to-understand manner.

C Expressed a warning to the other party through illustrations without using text.

D A list was compiled to summarize the events of a particular day.

(2) What is the term for symbols used in public facilities like train stations and airports to visually convey attention or information to others, as shown in the diagram on the right?



2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

([1]): A system for transferring information between humans and devices.
([2]): ([1]) through text. Operate everything from the keyboard.
([3]): ([1]) through visuals. Intuitive operation is possible using devices like a mouse.

A GUI **B** CUI **C** UX **D** UI

(2) Choose the most appropriate explanation of a signifier from the options **A** to **D** below, and answer using the letter.

A A measure used to indicate how easy and understandable it is for users.

B Design that is thoughtfully created with the assumption that everyone can use it without difficulty.

C A state that is easily accessible to a wide range of people.

D Cues that prompt users to take action.